

# Ahmed Ikram | Software Engineer

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Final-year BSc Computer Science (Data Science & AI), University of Dundee (predicted First, June 2027). Ship full-lifecycle software - design to deploy to real-user evaluation, fixing the issues users surfaced before deployment. Learned Terraform and cloud networking from zero inside an industry project. Work in standard PR and code-review conventions in university group project teams and stayed composed with difficult customers from front-of-house retail. Mentored junior pupils one-to-one as a classroom assistant. Seeking a graduate Software Engineering role from June 2027.

## EDUCATION

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**University of Dundee - BSc (Hons) Computer Science (Data Science and AI)** *Sep 2023 - Present*

- On track for a First-Class Honours degree. Expected Graduation: June 2027

**High School of Dundee** *Sep 2018 - June 2023*

- Highers: A Mathematics, A Computing Science, A English, B Chemistry
- Computing Science classroom assistant - mentored junior pupils one-to-one during classwork.

## TECHNICAL SKILLS

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**Languages:** Python · TypeScript · JavaScript · SQL · T-SQL · Rust / PyO3 · Bash

**Backend, APIs, Databases:** FastAPI · Flask · asyncio · GraphQL (Strawberry) · Pydantic · SQLAlchemy · Apache Kafka · Model Context Protocol (MCP) · PostgreSQL · SQLite · Redis · Alembic

**Frontend:** React · React Native · Expo · Tailwind CSS · Vite · Formik · GraphQL Codegen · graphql-ws · Socket.IO

**Security:** JWT · OAuth 2.0 · RBAC · bcrypt · AWS Secrets Manager

**Cloud & DevOps:** AWS (ECS Fargate, EC2, S3, RDS, ElastiCache, WAF, IAM, CloudFront, Secrets Manager, DynamoDB) · GCP (GKE, Cloud SQL, Cloud Monitoring) · Kubernetes (Kustomize, HPA, PDB, NetworkPolicy) · Azure (Databricks, Azure SQL, ADLS) · Terraform · Docker · Linux · GitHub Actions

**Observability & Testing:** Grafana · Prometheus · StatsD · Pytest · Vitest · Jest · Playwright · Cypress · k6 · React Testing Library

## PROJECTS

### DevSync Project Management Platform with GitHub Integration

*React · Tailwind · Flask · PostgreSQL · Socket.IO · GitHub OAuth · GCP · GKE · Terraform · GitHub Actions CI/CD*

- Realtime boards (tasks, live comments, presence) on JWT Socket.IO rooms that re-verify project membership per-event and degrade gracefully instead of crashing; Redis message queue fans out across replicas with cross-pod receipt tests.
- Zero-long-lived-credentials: GCP workload identity, digest-pinned images signed with cosign & SLSA L3 provenance & SBOM attestation, default-deny network policy, Terraform-managed cluster.
- Deployed to GKE: HPA on CPU, PDBs, private-IP Cloud SQL, ephemeral per-PR environments - running at ~£3-4/day with budget alerts.
- GitHub OAuth integration: issue/PR linking and comment write-back with encrypted token storage; 3-tier RBAC enforced server-side.
- 1,505 tests, with a k6 performance job gating every merge, p95 232ms vs a 500ms ceiling at 250/250 passing checks.

### ATM Log Aggregation, Anomaly Detection & Diagnostics (NCR Atleos)

*FastAPI · React · PostgreSQL · Kafka · Redis · Docker · Terraform · AWS · CI/CD*

- Architected a FastAPI REST API (30 endpoints, 6 routers, JWT + Redis-blacklist auth) with health-check-cascading startup across Kafka, Redis, and PostgreSQL.
- Built the Kafka (KRaft) event pipeline: at-least-once delivery, hybrid Redis/LRU deduplication, Redis Stream DLQ, sustaining ~100 msg/s over 2.5M events.
- Applied 8 Redis patterns (rate-limit, dedup, lock, pub/sub, cache, JWT blacklist, DLQ, counters) off one connection pool; React frontend ships as a ~25MB nginx-alpine image measured at build time.
- Provisioned Terraform across a multi-AZ VPC (ECS Fargate behind an ALB, an EC2 hosted Kafka & Redis in a private subnet); a 3-pipeline CI/CD runs 1,438 tests (959 pytest, 394 Vitest, 10 Playwright E2E, 75 Terraform assertions) and Checkov compliance on every PR, rolling out zero-downtime with zero long-lived credentials. State locked in DynamoDB.

## **StockLens FinTech Spending & Investment App**

*React Native · Expo · FastAPI · PostgreSQL · Redis · Rust · GraphQL · MCP · Terraform · Docker · AWS · CI/CD*

- Built an async FastAPI backend: 70+ REST endpoints with JWT auth (bcrypt), biometric unlock, and a stateless MCP server exposing 16 tools over streamable HTTP with OAuth 2.1 PKCE and JWT/JWKS validation.
- Replaced a pandas bottleneck with a native Rust extension (PyO3/Maturin, 12 functions) for technical-indicator computation, cross-built for ARM64 via QEMU in CI and shared by both the LSTM training pipeline and live API.
- Added a GraphQL read façade over the same service layer as REST: Strawberry resolvers with batched preloads eliminate N+1 queries, backed by a shared 60-second Redis quote cache; an authenticated graphql-ws market-data subscription replaces client polling.
- Deployed via Terraform (14 modules, 164 resources): three-tier network isolation (WAF-protected ALB → ECS Fargate ARM64 → private RDS/Redis) for a 20-30% compute-cost cut over x86; multi-stage images are 450MB vs 1.2GB naive.
- 1,584 backend and 844 frontend assertions gate a 9-job CI and an 8-job, 5-stage OIDC-authenticated CD pipeline.

## **W3C Web Logs ETL Pipeline**

*Airflow · Databricks · dbt · PySpark · Azure SQL · Terraform · Power BI · Grafana · Prometheus · Marquez · OpenLineage*

- Built a Databricks DLT medallion (Bronze → Silver in DLT; gold via dbt) on serverless distributed compute; 153,380 Bronze rows ingested with 0 dropped through 10 DLT quality gates (7 bronze & 3 Silver).
- 16 dbt models compile against PostgreSQL and T-SQL from one codebase via 18 compatibility macros; 121 dbt tests gate every model, catching 213 duplicate rows through a 12-component deduplication key.
- Integrated OpenLineage with a self-hosted Marquez collector across all 4 Airflow DAGs - task-run events, dbt model/column lineage, Azure SQL dataset edges, and custom data-quality facets in one cross-engine lineage graph.
- Deployed 30+ Azure resources through Terraform and OIDC federation; 627 tests, a freshness probe, 3 Grafana dashboards (23 panels), 8 Prometheus rules and 2 Azure Monitor alerts validate pipeline health and data quality.

## **CERTIFICATES**

AWS Academy Graduate - Machine Learning Foundations

*Feb 2025*

Foundations of Azure AI: Concepts, Capabilities, and Implementation

*Oct 2024*

AWS Academy Graduate - Cloud Foundations

*Mar 2024*

## **EXPERIENCE**

**Beaully Mini Market**, Dundee, Scotland

*Retail Assistant*

*Sep 2023 - Present*

**Zulus Peri Peri Grill**, Dundee, Scotland

*Front of House Worker*

*June 2021 - Sep 2023*